

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I **M. Meghana(192525435)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Meghana*

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Introduction

A Smart Retail Inventory Management System is a cloud-based solution that helps retail stores manage inventory, sales, and customer data efficiently. Store devices such as POS systems and dashboards connect to cloud services through APIs for real-time operations.

Components

- **Clients:** POS systems and dashboards send inventory and sales data.
- **API Gateway:** Manages communication and authentication.
- **Cloud Services (SaaS):** Inventory management and reporting.
- **Platform Services (PaaS):** Analytics and demand forecasting.
- **Cloud Storage:** Stores product details, transaction logs, and analytics data.
- **Infrastructure (IaaS):** Provides virtual servers, storage, and networking.

Security

- **VPN:** Secure communication between stores and cloud.
- **Firewall:** Protects against unauthorized access.
- **API Gateway:** Ensures secure API access and authentication.

Advantages

- Real-time inventory updates.
- Centralized data management.
- Accurate demand forecasting.

- Scalable and cost-effective.
- Improved security and business efficiency.

This cloud-based system combines **IaaS, PaaS, and SaaS** to provide secure, scalable, and efficient inventory management. It improves operational efficiency, supports better decision-making, and ensures reliable retail services.

Case Study 2: Cloud-Based Document Collaboration System

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Introduction

A Cloud-Based Document Collaboration System allows multiple users to create, edit, and share documents in real time through web browsers. It supports collaboration from different devices using cloud technology.

Components

Client Access

Users access the platform through browsers or mobile devices. The clients communicate with cloud services using Web APIs, which synchronize document updates across all connected devices in real time.

Cloud Storage

Documents are stored in distributed cloud storage, providing high availability, automatic backup, and scalability. A version control mechanism maintains different versions of documents, allowing users to restore previous copies if needed.

Cloud Services

The system uses Software as a Service (SaaS) for online document editing and sharing. Cloud infrastructure provides the required computing power, storage, and networking resources to support collaboration.

Network Infrastructure

The cloud network ensures low latency and high availability, enabling users around the world to access and edit documents without delays or interruptions.

Security

- Encryption: Protects documents during storage and transmission.
- Identity Management: Authenticates users.
- Access Permissions: Controls who can view or edit documents.

Advantages

- Real-time collaboration.
- Automatic document synchronization.
- Version control and backup.
- Secure document sharing.

The cloud-based document collaboration system uses **SaaS, cloud storage, and Web APIs** to enable secure and real-time document editing. It improves teamwork, ensures data security, and provides reliable access from anywhere, making collaboration efficient and flexible.

Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Introduction

An Online Food Ordering Platform is a cloud-based application that allows customers to browse menus, place food orders, and make online payments using web browsers or mobile applications. It provides fast, secure, and real-time food ordering services.

Components

Client Access

Customers access the platform through web browsers or mobile applications. The frontend communicates with backend services using RESTful Web APIs to retrieve menus, place orders, and process payments.

Cloud Storage

Cloud storage securely stores menu images, customer information, invoices, and order details. It provides reliable data backup, easy access, and scalability.

Cloud Services and Infrastructure

The application is hosted on a cloud platform using virtual machines and managed Kubernetes. Kubernetes automatically manages application deployment, scaling, and availability, ensuring reliable performance during high customer demand.

Security

Security is maintained through:

- Authentication Tokens to verify users.
- HTTPS to encrypt communication between clients and servers.
- Role-Based Access Control (RBAC) to restrict access based on user roles such as customer, restaurant, and administrator.

Advantages

- Real-time food ordering and tracking.
- Scalable and reliable cloud infrastructure.
- Secure data storage and communication.
- Easy access through web and mobile devices.
- High availability and improved customer experience.

The Online Food Ordering Platform integrates clients, RESTful Web APIs, cloud storage, virtual machines, and Kubernetes to deliver secure, scalable, and efficient food ordering services. It improves business operations while providing customers with a fast and reliable online ordering experience.